

Distinguishing Exponential from Linear Change

Algebra 1 Exponential Growth and Decay

Deciding from a table whether change is by a constant difference or a constant ratio, writing the matching rule, comparing a linear model with an exponential one, and classifying real situations before computing

The one idea behind every problem. Two patterns look alike for a step or two and then part company forever. Linear change **adds** the same amount each step; exponential change **multiplies** by the same factor each step. Every problem here asks the same first question: is the step an addition or a multiplication?

Linear: constant *difference* $f(n) = a + d \cdot n$

Exponential: constant *ratio* $f(n) = a \cdot r^n$

a is the starting value, d is what is added, r is what you multiply by

Reference card. No values are filled in — use the numbers given in each problem.

Directions.

- For every table, write the difference row *and* the ratio row before deciding. One of them will be constant, or neither will be.
- Say which model you chose and why, then compute.
- Round money answers to the nearest cent. Write your final answer on the answer line.

1. Here is a table of values for f .

n	0	1	2	3
$f(n)$	5	8	11	14

- (a) Write the differences between consecutive values, and the ratios. Which one is constant?
- (b) Is f linear or exponential? Write its rule.
- (c) Find $f(6)$.

Answer: _____

2. Here is a table of values for g .

n	0	1	2	3
$g(n)$	4	12	36	108

- (a) The differences here are 8, 24, 72 — not constant. Compute the ratios instead. What do you find?
- (b) Is g linear or exponential? Write its rule.
- (c) Find $g(5)$.

Answer: _____

3. Rosa opens a savings account with 200 dollars and adds 25 dollars at the end of every month. No interest is paid.

- (a) Is the balance changing by a constant difference or a constant ratio? Say which, in one sentence.
- (b) Write the rule for the balance after n months.
- (c) How many dollars are in the account after 8 months?

Answer: _____

4. A bacteria colony starts with 6 cells, and the number of cells doubles every hour.
- (a) “Doubles” names an operation. Is it addition or multiplication? What does that make the model?
 - (b) Write the rule for the number of cells after n hours.
 - (c) How many cells are there after 7 hours?

Answer: _____

5. Two savings plans both start at 100 dollars.
- **Plan L** adds 50 dollars each step.
 - **Plan E** multiplies by 1.5 each step.
- (a) Write a rule for each plan.
 - (b) Plan E is behind Plan L at step 1. Explain in one sentence how it can still end up ahead.
 - (c) At step 5, how many dollars is Plan E ahead of Plan L? Round to the nearest cent.

Answer: _____

6. Not every pattern is one of the two. Here is a table for h .

n	1	2	3	4
$h(n)$	2	8	18	32

- (a) Compute the first ratio $h(2) \div h(1)$.
(b) Compute the next ratio $h(3) \div h(2)$ as a decimal.
(c) The differences are 6, 10, 14 — not constant either. What do your two ratios prove about h ? Write both ratios on the answer line.

Answer: _____

7. A pattern is known to be linear, and two of its values are $f(2) = 17$ and $f(5) = 32$.
(a) What is the constant rate of change?
(b) Use it to write the rule for $f(n)$.
(c) If instead the pattern were exponential through those same two points, would the rate of change still be constant? Answer in one sentence.

Answer: _____

8. A car is worth 24000 dollars today. Two dealers describe how it will lose value.
- **Dealer A:** it loses 12 percent of its current value each year.
 - **Dealer B:** it loses 2400 dollars each year.
- (a) Which description is exponential and which is linear? How do you know?
- (b) Write a rule for each value after n years.
- (c) After 5 years, how many dollars more is the car worth under Dealer A than under Dealer B? Round to the nearest cent.

Answer: _____

9. A population grows exponentially. It is 500 at time 0 and 4000 at time 3.
- (a) Find the growth factor r per unit of time. Show the equation you solved.
- (b) Use your rule to find the population at time 5.
- (c) Write both answers on the answer line, in the order (a) then (b).

Answer: _____

10. Challenge. A summer job offers two pay plans for a 20-day stretch.

- **Flat plan:** 1000 dollars on every day.
- **Doubling plan:** 0.01 dollars on day 1, and the daily pay doubles each day after that.

(a) Write a rule for the doubling plan's pay on day n . Be careful with the exponent: day 1 has been doubled zero times.

(b) On day 10, which plan pays more that day? A one-line estimate is enough.

(c) On day 20, how many dollars more does the doubling plan pay that day than the flat plan? Round to the nearest cent.

Answer: _____